Selective Recovery for Multilingual ASR: Candidate Acquisition and Acceptance

Zhifan Pan[1,2,3], Maihemuti Maimaiti[1,2,3,*], Yingxiong Gao[1,2,3], Yuhui Ming[1,2,3], Aishan Wumaier[1,2,3]
[1]School of Computer Science and Technology, Xinjiang University, Urumqi, China
[2]Xinjiang Multimodal Intelligent Processing and Information Security
Engineering Technology Research Center, Xinjiang University, Urumqi, China
[3]Joint International Research Laboratory of Silk Road Multilingual
Cognitive Computing, Xinjiang University, Urumqi, China
*Corresponding author: mahmutjan@xju.edu.cn

***Abstract*—We separate fallback acquisition from acceptance using a compression trigger and a fixed four-feature selector. On a locked confirmation panel of 7,500 Common Voice 27.0 (CV27) recordings, selective recovery lowers macro word error rate (WER) by 2.26 percentage points (pp) over baseline rescoring and matches unconditional recovery's error counts in every seed. Retrospective CV27 TEST comparisons show 99.98% retention of macro-weighted positive edit gain. Reselection improves unscreened fallbacks; direct adoption skips trigger-time rescoring of screened Hugging Face (HF) outputs with comparable accuracy. Edit-gap refitting on HF banks yields the lowest exploratory CV27 WER of 31.44%, with checkpoint-specific gains. Across three checkpoints, selective acquisition saves up to 5.15% latency within reranked pipelines. Native HF remains faster but incurs 1.46 pp higher WER on FLEURS and over 2.1 pp on CV27. Code is available at https://github.com/Zhifan-Pan/ISCAS2027.**

***Index Terms*—multilingual speech recognition, selective recovery, candidate acquisition, N-best reranking, confidence estimation, conditional computation**

I. Introduction

Multilingual ASR encounters routine errors and catastrophic repetition loops. Additional decoding helps only when it produces a better candidate that the scoring rule accepts. Separating acquisition from acceptance reveals when recovery helps or wastes computation.

Multilingual N-best reranking [1], [2], discriminative rescoring [3]–[5], voting [6] and confusion networks [7] exploit existing hypotheses. Generative correction expands banks [8]–[11], and risk-based pruning and ASR sampling allocate computation [12], [13]. Conditional computation uses recognizer cascades [14], selective passes [15], encoder proxies [16], selective correction [17] or language-specific distillation [18], but confidence need not predict the benefit of deferral [19].

We distinguish routine beam rescoring from sparse repetition recovery. We ask when bank expansion helps and when reselection helps or harms a screened fallback. The locked panel isolates acquisition; retrospective controls examine acceptance. We compare expected risk, pairwise ranking and minimum Bayes risk on matched banks [20]–[22]. Native HF trades higher WER for faster decoding.

II. Controlled Acquisition and Acceptance

A. Recording-level acquisition

We normalize and deduplicate five beams per window i into B_i, retaining first occurrences and sources. With raw top beam b_i and fallback g_i, $F_i = B_i \cup \{g_i\}$ contains at most six candidates. Before recovery, $\mathrm{CR}_{\mathrm{text}}$ divides top-beam UTF-8 byte length by zlib-compressed length, with zero for empty strings. The gate $z_i = \mathbf{1}\{\mathrm{CR}_{\mathrm{text},i} > 2.4\}$ uses Whisper's standard text threshold [29], fixed before confirmation. Recording-level invocation $z^{\mathrm{rec}} = \max_i z_i$ expands every window, even singletons (Fig. 1), as a conservative control. On single-window CV27 audio, $z^{\mathrm{rec}} = z_i$; Section IV-D examines local routing for multi-window FLEURS.

Recovery follows Hugging Face temperatures $0, 0.2, \ldots, 1$ with compression, log-probability and no-speech thresholds 1.35, -1 and 0.6. The internal check compresses serialized token-ID bytes [23] with a stricter retry threshold; the outer gate screens decoded text for severe loops. Shared leading bytes in two-byte Arabic/Persian characters can increase compression relative to ASCII Latin text at similar character repetition. Native HF uses the same retry configuration without N-best rescoring.

B. Four-feature shared acceptance

For features $f(y) = (a, p, t, r)$, a is the mean teacher-forced log probability over T body tokens without prefix, padding or end tokens, and $t = \log(1 + T)$. XLM-R-base pseudo-log-likelihood p [24], [25], or PLL, averages individually masked nonspecial subword log probabilities in centered contexts of at most 510 subwords of the original string, rather than within-word masking [26]. Empty outputs have no body tokens or PLL targets and map to $f(\emptyset) = (0, 0, 0, 0)$, but normalization alone need not yield zero features.

For normalized words $w_{1:M}$, the sequence-wide overlapping count is $c_y(u) = \sum_{k=1}^{M-n+1} \mathbf{1}\{w_{k:k+n-1} = u\}$. We measure repetition by

$$r_{n,h}(y) = \frac{\sum_{j=1}^{M-n+1} \mathbf{1}\{c_y(w_{j:j+n-1}) \geq h\}}{\max(1, M - n + 1)}. \quad (1)$$

We use $r(y) = r_{4,3}(y)$, which is zero for $M < 4$. Repetitive sequences also motivate unlikelihood training [27].

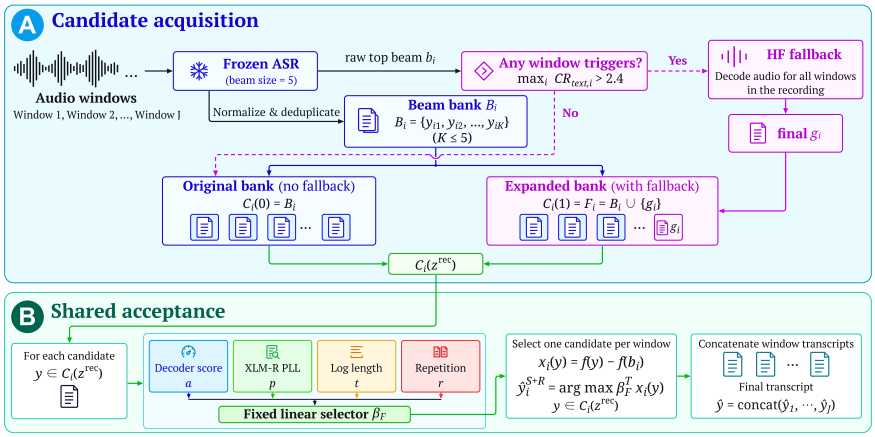


Fig. 1. Acquisition (A) expands every window's candidate bank if any window triggers. Acceptance (B) applies the same fixed four-feature selector to the original or expanded bank. Selected window transcripts are then concatenated.

Centering $x_i(y) = f(y) - f(b_i)$ preserves rankings and softmax probabilities. Under fitted signs $(+,+,-,-)$, $\beta_F^\top f(y) < 0 = \beta_F^\top f(\emptyset)$ for every nonempty y, so any HF-proposed empty wins deterministically. For $x_{ik} = x_i(y_{ik})$, root-mean-square (RMS) scales satisfy $s_d^2 = \sum_{\mathcal{D}} \omega_{ik} x_{ik,d}^2 / \sum_{\mathcal{D}} \omega_{ik}$ over non-top training-fold pairs $\mathcal{D} = \{(i,k) : i \in \bigcup_{\ell=1}^{L} \mathcal{I}_\ell,\ y_{ik} \in C_i \setminus \{b_i\}\}$. G4 uses $\omega_{ik} = 1/[N_\ell(|C_i| - 1)]$; all HF refits use $\omega_{ik} = 1$. Scales below 10^{-12} become one. For reference y_i^* and word-edit count E, define $\Delta_{ik} = E(b_i, y_i^*) - E(y_{ik}, y_i^*)$ and $\pi_{ik} = \text{softmax}_k\left(\theta^\top \text{diag}(s)^{-1} x_{ik}\right)$. Both softmax and loss include all candidates, with $x_i(b_i) = \mathbf{0}$ and $\Delta_{i,b_i} = 0$. Expected-risk fitting [20] minimizes

$$\mathcal{L}(\theta) = -\frac{1}{L} \sum_{\ell=1}^{L} \frac{\sum_{i \in \mathcal{I}_\ell} \sum_{k=1}^{|C_i|} \pi_{ik} \Delta_{ik}}{N_\ell} + \lambda \|\theta\|_2^2. \quad (2)$$

Here $\mathcal{I}_\ell = \{i : \text{lang}(i) = \ell\}$ indexes fitting windows, N_ℓ counts reference words and $L = 5$.

The shared weights $\beta_F = \text{diag}(s)^{-1}\theta$ are fitted per run on greedy-expanded DEV and fixed across sources, with ties broken by bank order. With $\hat{y}_i^B = \text{argmax}_{y \in B_i} \beta_F^\top x_i(y)$, $C_i(0) = B_i$ and $C_i(1) = F_i$, the policies are

$$\hat{y}_i^{S+R} = \underset{y \in C_i(z^{\text{rec}})}{\text{argmax}}\ \beta_F^\top x_i(y),$$
$$\hat{y}_i^{\text{Direct}} = \begin{cases} \hat{y}_i^B, & z^{\text{rec}} = 0, \\ g_i, & z^{\text{rec}} = 1. \end{cases} \quad (3)$$

Policy B reranks B_i, and N+R requests and reranks F_i unconditionally. S+R and Direct share acquisition and use B off-trigger, but Direct bypasses selection and PLL on triggers. Window transcripts are concatenated for full-reference scoring.

C. Matched acceptance controls

G4 stays frozen. We fit H4 by expected risk on HF-expanded DEV banks and H3 without PLL, using (a, t, r). RankNet [21] minimizes

$$\mathcal{L}_{\text{rank}} = \frac{1}{L} \sum_{\ell=1}^{L} \frac{\sum_{i \in \mathcal{I}_\ell} \ell_i}{N_\ell} + \lambda \|\theta\|_2^2,$$

with $\ell_i = |\mathcal{P}_i|^{-1} \sum_{(j,k) \in \mathcal{P}_i} w_{jk} \log(1 + e^{-\beta^\top (x_{ij} - x_{ik})})$ over unequal-edit pairs $\mathcal{P}_i = \{(j,k) : E_j < E_k\}$, where $E_j = E(y_{ij}, y_i^*)$ and $\beta = \text{diag}(s)^{-1}\theta$. Empty $\mathcal{P}_i$ give $\ell_i = 0$. HR4 uses $w_{jk} = 1$; HWR4 uses $|E_j - E_k|/c$, with c the training-fold mean gap under HR4's pair weights.

TABLE I
WER (%) AND PP CONTRASTS BEFORE ROUNDING. CHECKS DENOTE HF ACQUISITION, CR GATING AND R RESELECTION (†: OFF-TRIGGER ONLY).

Path	HF	CR	R	Scorer	FLEURS	CV27
Raw beam-5 top-1	–	–	–	–	18.90	36.41
Native HF [23]	✓	–	–	–	18.55	33.80
B	–	–	✓	G4	17.45	33.98
N+R	✓	–	✓	G4	17.09	31.68
S+R	✓	✓	✓	G4	17.09	31.68
Direct	✓	✓	✓†	G4	17.09	31.46
H3Direct	✓	✓	✓†	H3	18.68	33.64
A. Reference-assisted TEST oracle bounds						
Original-bank oracle					16.19	30.19
Full-HF oracle					15.83	27.68
CR-routed oracle					15.83	27.68
S+R gap to routed oracle, pp					1.26	4.00

B. Locked CV27 confirmation (7,500 recordings)			
	B	N+R = S+R	Gain
Macro-average	35.02	32.75	2.26
Contrast		ΔWER	95% upper bound
S+R − B		-2.26	-1.32
S+R − N+R		0.00	0.00

MBR [22] selects $\hat{y}_i = \text{argmin}_{y \in C_i} \sum_{y' \in C_i} q_i(y') E(y, y')$. U-MBR uses $q_i(y') \propto 1$ and D-MBR uses $q_i(y') = \text{softmax}(Ta/\tau)_{y'}$. DEV WER selects $\tau \in \{0.1, 0.3, 1, 3, 10\}$, favoring larger values on ties; candidate ties retain bank order. *Complete* policies use each acceptor on both banks; *triggered* comparisons keep G4 off-trigger with full-panel denominators.

III. Data and Evaluation

We adapt Whisper-small [29] on 44.60 h of FLEURS TRAIN [28] across Arabic, Persian, Polish, Turkish and Italian using LoRA [30] over three random seeds (4,656 updates, rank and scale four, zero dropout across all attention projections).

LoRA sees no CV27 data, so CV27 tests zero-shot acoustic transfer across domains. Selectors do use CV27 DEV, where recovery was checked with the fixed gate before confirmation. Evaluation on FLEURS TEST (13.07 h; 3,665 recordings) and CV27 TEST (83.00 h; 58,514 recordings) [31] is retrospective because both splits were accessed during development.

The locked confirmation panel has 7,500 CV27 TRAIN recordings (1,500 per language) with no previously accessed audio or reference texts. Predictions, stopping rules and the one-sided 0.10 pp noninferiority margin were locked before reference evaluation. We use 10,000 synchronized bootstrap draws of speaker components within languages. Speakers recurring across splits limit generalization to this sampling frame. Confirmation covers the original G4 acquisition policy; later gates, refits and window policies remain retrospective.

We weight costs by inverse sampling probabilities for 255 FLEURS and 472 CV27 recordings. The sample covers all first-checkpoint triggers or Native HF output changes and ten ordinary recordings per language and duration quintile. Study A measures four paths across three checkpoints; Study B measures seven on one. Later-checkpoint omissions comprise five FLEURS and 207 CV27 triggers (0.05% and 0.12% of seed–recordings). Triggered N+R and S+R execute identical operations, though sampling shares do not bound timing variation. Study B covers all its triggers and corroborates Study A's savings (5.18% and 4.54% versus 5.15% and 4.57%). Both use three randomized paired rounds with each recording block on one of four NVIDIA A40 (48 GB) GPUs, resident models, within-path encoder and first-search reuse, and 32-mask PLL batches. Waveform-to-output timing excludes loading, warmup and interfered blocks. Peak CUDA memory is the maximum per-process allocation in MiB over calls and rounds. Session-conditional intervals use 4,000 paired stratified draws with finite-population correction; Study A averages checkpoints with synchronized draws.

Known-language decoding uses bfloat16, five beams, length penalty one, native suppression and a 444-token cap in Transformers 4.57.6. At 16 kHz, N samples give $J = \lceil N/480000 \rceil$ windows bounded by $\lfloor jN/J \rfloor$ without previous-window text. Deduplication, WER and r share NFC normalization, digit unification, punctuation-to-space, Persian mapping and Turkish casing [32], with fixed source RNG mappings. WER averages five language aggregates and three seeds; calls pool seed–recordings. TEST intervals use 2,000 paired client/reference-component draws within languages, synchronized across seeds with fits and source draws frozen. FLEURS comparisons are descriptive because component metadata are unverified.

IV. Results

A. Acquisition with fixed acceptance

On the locked confirmation panel, S+R gains 2.26 pp over B, with seed gains of 0.20 to 4.54 pp (Table I). Matching N+R's error counts in every seed meets the prespecified 0.10 pp margin but does not guarantee zero loss under distribution shift. Of 61 triggers, 49 improve edits and two worsen them.

TABLE II
Complete-policy TEST WER (%) and gain B−S (pp) on HF banks. Checks denote PLL, pairwise P and edit-gap weighting W.

Acceptor	PLL	P	W	FLEURS			CV27		
				B	S	Gain	B	S	Gain
G4 [20]	✓	–	–	17.45	17.09	0.37	33.98	31.68	2.30
H4 [20]	✓	–	–	17.46	17.09	0.37	33.97	31.52	2.45
HR4 [21]	✓	✓	–	17.51	17.14	0.37	34.02	32.30	1.72
H3 [20]	–	–	–	19.05	18.68	0.37	36.15	33.77	2.38
HWR4 [21]	✓	✓	✓	17.47	17.10	0.37	33.97	31.44	2.53
U-MBR [22]	–	–	–	19.14	19.14	0.00	36.91	36.88	0.03
D-MBR [22]	–	–	–	18.92	18.75	0.17	36.09	34.48	1.60

A. Original family: CV27 ΔWER (individual 97.5% CI)		
Triggered HR4 − H4	+0.72	[+0.40, +1.16]
Complete H3 − H4	+2.25	[+2.16, +2.34]
B. Later two-comparison family		
Triggered U-MBR − H4	+2.51	[+1.93, +3.18]
Triggered HWR4 − H4	-0.08	[-0.15, -0.03]

Retrospective evaluation yields 32.71% WER for Direct versus S+R's locked 32.75%. Both beat B and Native HF's retrospective 34.87% by over 2.1 pp. Direct's unrounded difference from S+R is -0.05 pp with a paired speaker-bootstrap 95% interval from -0.19 to $+0.04$ pp.

On TEST, S+R retains all ten beneficial FLEURS instances at 0.09% calls and 370 of 373 CV27 instances at 0.30%, missing five edits. Language-reference normalization gives 99.98% macro-weighted positive edit gain retention. N+R selects HF on one FLEURS and 13 CV27 nontriggers where S+R keeps B, with nearly identical WER. Oracles minimize edits over concatenated original, expanded or routed banks. Routing changes this bound by under 0.01 pp against S+R's 4.00 pp CV27 gap. In retrospective replay on FLEURS/CV27 banks, $r_{4,3}(b_i) > 0$ gives 0.06%/0.18% calls and 93.34%/99.21% retention. Token-ID byte compression > 1.35 gives 0.20%/0.31% calls and 100%/99.99% retention.

B. Interaction with the fallback source

S+R beats Direct by 0.53 pp with unscreened greedy recovery and 0.43 pp at temperature 0.2. HF screens retries by compression, log-probability and no-speech criteria. Direct then leads G4 by 0.22 pp, with a 95% interval from 0.11 to 0.37 pp, concentrated in checkpoint one (-0.68 pp versus $+0.00$ and $+0.02$ pp). G4 retains repetitive beams whose decoder/PLL advantage outweighs length and repetition penalties. DEV favored reselection at 30.08% versus 30.09%. Thus Direct simplifies trigger-time scoring with comparable accuracy, rather than consistently improving it.

Direct retains 67.13% of positive gain from unconditional final HF, compared with S+R's 99.98% from N+R. Relative to its own control, Direct forgoes 1.24 pp of benefit but avoids 3.58 pp of harm.

C. Acceptance on matched HF banks

On HF banks, H4 improves over G4 but trails Direct (31.52% versus 31.46% CV27 WER). HWR4 yields the lowest complete CV27 WER of 31.44% (Table II). Triggered HR4 trails H4 in every seed ($+0.91$, $+0.08$ and $+1.17$ pp). Edit-gap

TABLE III
POPULATION-WEIGHTED MS AND OBSERVED PEAK ALLOCATED MIB WITH BOTH MODELS RESIDENT. EACH STUDY USES THREE PAIRED ROUNDS.

Path	FLEURS		CV27	
	ms	MiB	ms	MiB
A. Four-path comparison across three checkpoints				
Native HF	1291.7	2980.8	517.4	2813.8
B	1352.7	3144.9	591.9	3133.1
N+R	1436.8	2988.3	639.2	3133.1
S+R	1362.8	3189.7	610.0	3133.1
B. Seven-path comparison on one checkpoint				
Native HF	1304.8	2978.4	531.2	2811.8
Raw top-1	1222.6	2946.6	479.9	2571.0
B	1364.0	3142.4	605.1	3131.1
N+R	1449.6	2986.3	657.9	3131.1
S+R	1374.5	3187.2	628.0	3131.1
Direct	1372.8	3187.2	625.8	2811.8
H3Direct	1314.0	3187.2	586.4	2811.8
Relative latency saving: S+R versus N+R (95% CI)				
Study A	5.15% [2.84, 9.41]		4.57% [4.40, 4.73]	
Study B	5.18% [2.88, 9.39]		4.54% [4.44, 4.66]	
Study B: differences between policies				
B − Raw top-1, ms	141.4		125.2	
N+R − B, ms	85.6		52.8	
S+R − B, ms	10.5		22.9	
Fallback increment saved	87.7%		56.6%	

weighting reverses the mean contrast to -0.08 pp. Its 0.25 pp gain in checkpoint two replaces 11 repetitive beams with lower-edit HF candidates. Intervals provide 95% Bonferroni coverage per family, with 10,000 synchronized draws for the later family. Repeating DEV selection with G4's RMS rule changes complete CV27 WER by $+0.044$, -0.057 and $+0.003$ pp for H4, HR4 and HWR4, respectively. Their ordering is unchanged.

FLEURS gains round to 0.37 pp for fitted acceptors because S equals its own B off-trigger. H3's B has two fewer errors on one of ten triggers, but S errors match and the average difference is below 0.006 pp. Nontriggers account for 97.23% of the refitted H3–H4 WER gap in routine selection, without isolating PLL's causal contribution or time share.

D. Cost, heterogeneity and supporting diagnostics

Within reranked pipelines, S+R preserves N+R's gain at lower cost (Table III). Study B's rescoring increments above raw top-1 are 141.4 ms and 125.2 ms on FLEURS and CV27. Selective acquisition saves 87.7% and 56.6% of fallback overhead. The unweighted 25-recording CV27 profile averages 85.9 ms teacher scoring and 149.3 ms PLL, including two skipped singletons. More singletons skip both stages in the population, explaining the smaller weighted increments. Native HF skips rescoring, trading faster decoding for 1.46 pp higher FLEURS WER and over 2.1 pp on CV27 TEST.

H3Direct uses H3 off-trigger and HF on triggers, omitting PLL. On CV27 it saves 6.29% over Direct but trails Native HF by 55.2 ms (95% interval 52.3 to 58.2 ms); the FLEURS interval spans zero. H4 and HWR4 only replace G4's weights; selective PLL remains future work. In Study B, Direct skips scoring on the CV27 trigger responsible for B's peak CUDA memory. On FLEURS, Direct and H3Direct match S+R's peak CUDA allocation when both windows' states are retained during recovery.

TABLE IV
TEST B WER (%) AND GAIN B−(S+R) (PP), AVERAGED OVER SEEDS BEFORE ROUNDING. CALLS POOL SEED–RECORDINGS.

Language	FLEURS			CV27		
	B	Gain	Calls, %	B	Gain	Calls, %
Arabic	24.84	0.64	0.23	52.12	1.40	0.11
Persian	31.08	1.20	0.27	69.02	9.05	1.15
Polish	13.57	0.00	0.00	14.57	0.30	0.33
Turkish	11.67	0.00	0.00	20.52	0.51	0.03
Italian	6.11	0.00	0.00	13.66	0.23	0.02
Empty-HF acceptance diagnostic (18 instances, 11 recordings)						
Acceptance minus B reversion	$S = -118$		$D = +120$		$I = -1995$	

Only HF proposes empty TEST candidates. Since nonempty features satisfy $a, p \leq 0$, $t > 0$ and $r \geq 0$, the intercept-free rule deterministically passes empties through, without independently validating abstention. The 18 acceptances in 11 CV27 recordings remove 1,995 insertions and add 120 deletions (Table IV), consistent with hallucination suppression [33] but without verified silence labels. Reverting all to B retains 91.52% of gain, or 2.10 pp at 31.88% WER, so recovery mainly reflects nonempty repairs. Refitting H4 with an empty-specific feature $e(y) = \mathbf{1}\{y = \emptyset\}$ retains all 18 acceptances at 31.52% WER. One or two DEV empties per checkpoint limit calibration.

Cross-domain CV27 B WER reaches 69.02% for Persian (71.61% raw) and 52.12% for Arabic. Acoustic and orthographic mismatch may contribute, but their effects are not isolated. Persian supplies 78.77% of TEST recovery gain and 84.07% on the locked panel. Excluding Persian, four-language macro gains remain 0.61 pp on CV27 TEST and 0.45 pp on the locked panel. The 22 two-window FLEURS recordings yield 52.69% of net gain. On cached TEST banks, local S+R reduces window requests from 15 to 10 without changing errors. Local Direct changes WER by under 0.01 pp, though live request order may alter RNG draws.

V. CONCLUSION

On the locked panel, selective acquisition matches unconditional recovery across seeds. Separate timing studies show lower recovery cost. Retrospective controls favor expected risk over unweighted ranking on matched HF banks in every seed.

Acceptance should reflect fallback screening. Direct skips trigger-time rescoring of screened HF outputs with comparable accuracy; unscreened draws benefit from reselection. HWR4 has the lowest exploratory CV27 WER, but its advantage over H4 is concentrated in one checkpoint. Future work includes script-normalized gates, window-local routing and confidence-gated PLL.

ACKNOWLEDGMENT

Supported by the National Natural Science Foundation of China under Grant 62466058 and Autonomous Region Key R&D Program under Grant 2025B03042-1.

References

[1] N. Gaur, T. Chen, E. Variani, P. Haghani, B. Ramabhadran, and P. J. Moreno, "Multilingual second-pass rescoring for automatic speech recognition systems," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 2022, pp. 6407–6411.

[2] B. Yan, V. Pratap, S. Watanabe, and M. Auli, "Improving multilingual ASR in the wild using simple N-best re-ranking," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 2025, pp. 1–5.

[3] L. Xu *et al.*, "RescoreBERT: Discriminative speech recognition rescoring with BERT," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 2022, pp. 6117–6121.

[4] P. G. Shivakumar, J. Kolehmainen, Y. Gu, A. Gandhe, A. Rastrow, and I. Bulyko, "Discriminative speech recognition rescoring with pre-trained language models," in *Proceedings of the IEEE Automatic Speech Recognition and Understanding Workshop*, 2023, pp. 1–7.

[5] A. Gandhe and A. Rastrow, "Audio-attention discriminative language model for ASR rescoring," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 2020, pp. 7944–7948.

[6] J. G. Fiscus, "A post-processing system to yield reduced word error rates: Recognizer output voting error reduction (ROVER)," in *Proceedings of the IEEE Automatic Speech Recognition and Understanding Workshop*, 1997, pp. 347–354.

[7] L. Mangu, E. Brill, and A. Stolcke, "Finding consensus in speech recognition: Word error minimization and other applications of confusion networks," *Computer Speech and Language*, vol. 14, no. 4, pp. 373–400, 2000.

[8] C.-H. Kuo and K.-Y. Chen, "An integrated framework for ASR post-processing with error correction, rescoring, and text-speech matching," *APSIPA Transactions on Signal and Information Processing*, vol. 15, no. 1, pp. 223–248, 2026, doi: 10.1108/ATSIP-07-2025-0069.

[9] S. Dutta, S. Jain, A. Maheshwari, S. Pal, G. Ramakrishnan, and P. Jyothi, "Error correction in ASR using sequence-to-sequence models," arXiv:2202.01157, 2022.

[10] A. D. Tur, A. Moumen, and M. Ravanelli, "ProGRes: Prompted generative rescoring on ASR N-best," in *Proceedings of the IEEE Spoken Language Technology Workshop*, 2024, pp. 600–607, doi: 10.1109/slt61566.2024.10832194.

[11] S. Radhakrishnan, C.-H. Yang, S. Khan, R. Kumar, N. Kiani, D. Gomez-Cabrero, and J. Tegnér, "Whispering LLaMA: A cross-modal generative error correction framework for speech recognition," in *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, 2023, pp. 10007–10016, doi: 10.18653/v1/2023.emnlp-main.618.

[12] J. Cheng and A. Vlachos, "Faster minimum Bayes risk decoding with confidence-based pruning," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023, pp. 12473–12480, doi: 10.18653/v1/2023.emnlp-main.767.

[13] Y. Jinnai, "Re-evaluating minimum Bayes risk decoding for automated speech recognition tasks," *Transactions on Machine Learning Research*, 2026. [Online]. Available: https://openreview.net/forum?id=I6iLWhRIsf.

[14] H. Xue, Y. Tang, J. Zhang, X. Geng, and L. Xie, "Selective invocation for multilingual ASR: A cost-effective approach adapting to speech recognition difficulty," in *Proceedings of Interspeech*, 2025, pp. 2580–2584, doi: 10.21437/interspeech.2025-1652.

[15] P. Aleksic and L. Zhou, "Latency reduction in conversational user interfaces by selective execution of multi-pass speech recognition based on LLM evaluation," *Technical Disclosure Commons*, technical disclosure, Jun. 27, 2025. [Online]. Available: https://www.tdcommons.org/dpubs_series/8287/.

[16] Y. Fathullah, P. Radmard, A. Liusie, and M. Gales, "Who needs decoders? Efficient estimation of sequence-level attributes with proxies," in *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics*, 2024, pp. 1478–1496, doi: 10.18653/v1/2024.eacl-long.89.

[17] Y. Lim, J. Park, H. Y. Kim, H. Lee, and B.-Y. Kim, "Hybrid decoding: Rapid pass and selective detailed correction for sequence models," in *Proceedings of the IEEE Automatic Speech Recognition and Understanding Workshop*, 2025, pp. 1–7, doi: 10.1109/ASRU65441.2025.11434678.

[18] T. P. Ferraz, M. Zanon Boito, C. Brun, and V. Nikoulina, "Multilingual DistilWhisper: Efficient distillation of multi-task speech models via language-specific experts," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 2024, pp. 10716–10720, doi: 10.1109/ICASSP48485.2024.10447520.

[19] W. Jitkrittum, N. Gupta, A. K. Menon, H. Narasimhan, A. S. Rawat, and S. Kumar, "When does confidence-based cascade deferral suffice?," in *Advances in Neural Information Processing Systems*, vol. 36, 2023, doi: 10.52202/075280-0431.

[20] R. Prabhavalkar *et al.*, "Minimum word error rate training for attention-based sequence-to-sequence models," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 2018, pp. 4839–4843.

[21] C. Burges *et al.*, "Learning to rank using gradient descent," in *Proceedings of the 22nd International Conference on Machine Learning*, 2005, pp. 89–96.

[22] V. Goel and W. J. Byrne, "Minimum Bayes-risk automatic speech recognition," *Computer Speech and Language*, vol. 14, no. 2, pp. 115–135, Apr. 2000.

[23] Hugging Face, "Whisper generation implementation," *Transformers*, v4.57.6, documentation and `_retrieve_compression_ratio`. [Online]. Available: https://github.com/huggingface/transformers/blob/v4.57.6/src/transformers/models/whisper/generation_whisper.py

[24] A. Conneau *et al.*, "Unsupervised cross-lingual representation learning at scale," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2020, pp. 8440–8451.

[25] J. Salazar, D. Liang, T. Q. Nguyen, and K. Kirchhoff, "Masked language model scoring," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2020, pp. 2699–2712.

[26] C. Kauf and A. Ivanova, "A better way to do masked language model scoring," in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics, Vol. 2: Short Papers*, 2023, pp. 925–935.

[27] S. Welleck, I. Kulikov, S. Roller, E. Dinan, K. Cho, and J. Weston, "Neural text generation with unlikelihood training," in *Proceedings of the International Conference on Learning Representations*, 2020.

[28] A. Conneau *et al.*, "FLEURS: Few-shot learning evaluation of universal representations of speech," in *Proceedings of the IEEE Spoken Language Technology Workshop*, 2023, pp. 798–805. [Online]. Available: https://huggingface.co/datasets/google/fleurs.

[29] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 202, 2023, pp. 28 492–28 518. [Online]. Available: https://proceedings.mlr.press/v202/radford23a.html.

[30] E. J. Hu *et al.*, "LoRA: Low-rank adaptation of large language models," in *Proceedings of the International Conference on Learning Representations*, 2022. [Online]. Available: https://openreview.net/forum?id=nZeVKeeFYf9.

[31] R. Ardila *et al.*, "Common Voice: A massively-multilingual speech corpus," in *Proceedings of the Twelfth Language Resources and Evaluation Conference*, 2020, pp. 4218–4222. [Online]. Available: https://aclanthology.org/2020.lrec-1.520/.

[32] K. Manohar and L. G. Pillai, "What is lost in normalization? Exploring pitfalls in multilingual ASR model evaluations," in *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, 2024, pp. 10 864–10 869.

[33] M. Barański, J. Jasiński, J. Bartolewska, S. Kacprzak, M. Witkowski, and K. Kowalczyk, "Investigation of Whisper ASR hallucinations induced by non-speech audio," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 2025, pp. 1–5, doi: 10.1109/ICASSP49660.2025.10890105.